SYNOPSYS ACQUIRES ANSYS ($35.0 BILLION)

1. Executive Thesis and Transaction Snapshot

On 16 January 2024, Synopsys agreed to acquire Ansys for approximately US$35 billion in cash and stock, marking the largest technology deal announced that year. After an eighteen-month review by regulators in the United States, Europe, the United Kingdom, and China, the transaction closed on 17 July 2025.

The strategic case was clear. Chip design and physical simulation had long operated as separate disciplines. That separation stopped making sense as AI processors moved into advanced 3D packaging and more companies began designing their own silicon. At these densities, heat, power, and electromagnetic effects directly affect how the chip performs, so customers increasingly needed one connected workflow rather than two disconnected toolsets. By combining Ansys's simulation capabilities with Synopsys's EDA platform, the deal expands the addressable market by roughly 1.5 times to between US$28 billion and US$31 billion and creates a true silicon-to-systems offering.

What makes the transaction stand out from an M&A perspective is the discipline with which it was structured. Synopsys used a balanced mix of cash and stock, placed a hard 19.9999 per cent cap on new share issuance to avoid a buyer shareholder vote, and proactively offered targeted divestitures to clear antitrust reviews. On the financing side, it raised US$14.3 billion of investment-grade debt while committing to bring leverage back below 2.0 times within two years. The premium was underwritten primarily on long-term cross-selling and platform integration rather than aggressive cost cutting.

Transaction Parameter	Disclosed Terms
Acquirer / Target	Synopsys (SNPS) / Ansys (ANSS)
Announcement / Closing	Announced 16 January 2024; Closed 17 July 2025
Offer Price & Premium	US$390.19 per share (29% unaffected premium)
Enterprise Value	Approximately US$35.0 billion
Implied Multiples	15.4x trailing revenue; >35.0x trailing adjusted EBITDA
Consideration Mix	Announced: US$197.00 cash + 0.3450 SNPS shares; Closed: US$199.91 cash + 0.3399 shares (19.9999% share cap)
Pro Forma Ownership	Approximately 83.5% Synopsys / 16.5% Ansys
Financing	US$10.0 billion senior notes + US$4.3 billion term loan + cash (including US$2.1 billion Software Integrity divestiture)
Synergies	US$400 million cost by Year 3; US$400 million revenue by Year 4 (scaling above US$1.0 billion long-term)
Termination Fees	US$1.5 billion reverse fee (Synopsys); US$950 million target fee
Advisers	Evercore and Cleary Gottlieb (Synopsys); Qatalyst Partners (Ansys)

Source: Synopsys Form S-4 Registration Statement, Merger Agreement, and Closing Form 8-K Filings.

2. Strategic Rationale and Value-Creation Levers

Chip architects historically used EDA tools from Synopsys or its main rival Cadence Design Systems to design circuits and verify timing, while mechanical engineers used Ansys software to test heat, airflow, and structural stress on finished products. Running those checks separately worked when chips were simpler, but advanced packaging has linked electrical and physical performance directly together.

Three industry trends accelerated the need to combine both toolsets. First, AI chips stack multiple dies into three-dimensional packages where localized heat and electromagnetic interference directly alter circuit speed. Second, automotive and industrial manufacturers building custom silicon want a single engineering workflow from the chip to the finished system. Third, both companies sell mission-critical software on multi-year subscription contracts with high renewal rates, giving the combined business durable recurring cash flow.

Management anchored the acquisition price on two synergy targets. On the cost side, Synopsys targeted US$400 million in annual run-rate savings by Year 3 by removing duplicate public-company costs, combining overlapping sales teams, and streamlining vendor spend. On the revenue side, management underwrote US$400 million in annual cross-selling synergies by Year 4, scaling above US$1.0 billion over time as Ansys simulation tools are sold into Synopsys's semiconductor accounts and bundled for automotive and aerospace clients. Offering a unified cloud-ready workflow also shortens customer design cycles and strengthens pricing power against Cadence and standalone simulation peers.

3. Valuation Framing and Financial Accretion

Per the joint proxy statement and Form S-4 filings, the offer price of US$390.19 per share valued Ansys at roughly US$35.0 billion against trailing twelve-month revenue of US$2.27 billion. Because Ansys held net cash and generated gross margins above 85 per cent, investors evaluated the purchase price primarily on revenue multiples, free cash flow conversion, and multi-year earnings accretion rather than near-term GAAP multiples.

Paying a 29 per cent unaffected control premium placed the deal toward the upper end of large design-software takeovers, which have historically cleared at 12 to 16 times revenue, and above unaffected engineering software peers trading at 10 to 14 times forward revenue. Synopsys justified that premium through Ansys's above-85 per cent gross margins, sticky subscription base, and the 1.5 times expansion in addressable market size.

Valuation Benchmark	Period	EV / Revenue	EV / EBITDA	Market Context
Synopsys / Ansys Transaction	2024/25	15.4x	>35.0x	Strategic control; US$35.0B EV
Large Design-Software Takeovers	Precedent	12.0x–16.0x	N/A	Control range at announcement
Engineering Software Peers	Pre-Deal	10.0x–14.0x	N/A	Unaffected public trading range

Source: Synopsys Form S-4 Registration Statement, Merger Announcement Presentation, and Public Market Disclosures.

Management projected that combining the two companies would lift non-GAAP operating margins by roughly 125 basis points and unlevered free cash flow margins by about 75 basis points in the first full year after closing. While higher interest expense kept Year 1 non-GAAP earnings per share roughly neutral to modestly dilutive, the deal turns meaningfully accretive in Year 2 once the bulk of the US$400 million cost synergies is captured.

4. Deal Process, Financing Structure, and Credit Impact

Negotiations focused on four structural choices: balancing cash and stock consideration, capping equity issuance below NASDAQ's 20 per cent shareholder-vote threshold, sizing the reverse termination fee for regulatory risk, and securing committed debt without losing Synopsys's investment-grade rating. Evercore and Cleary Gottlieb advised Synopsys on protecting governance and balance-sheet flexibility, while Qatalyst Partners advised Ansys on securing the 29 per cent control premium, a US$1.5 billion reverse termination fee, and a long outside date to accommodate antitrust reviews.

Under the merger agreement's share-issuance cap, total new stock could not exceed 19.9999 per cent of Synopsys's pre-closing share count, leaving Synopsys shareholders with roughly 83.5 per cent pro forma ownership. Because of that cap, the headline mix of US$197.00 in cash and 0.3450 Synopsys shares per Ansys share adjusted automatically at closing on 17 July 2025 to US$199.91 in cash and 0.3399 Synopsys shares.

Disclosed Cash & Stock Sources	Amount (US$B)	Disclosed Transaction Uses	Amount (US$B)
Senior Unsecured Notes (March 2025)	$10.0	Cash Consideration to Ansys Shareholders	$19.0
Senior Unsecured Term Loan Drawn at Close	$4.3	Stock Consideration (Share-Cap Adjusted)	$16.0
Synopsys Balance-Sheet Cash (incl. Divestiture Proceeds)	$4.7		
Total Disclosed Funding Sources	**$35.0**	**Headline Enterprise Value**	**$35.0**

Source: Synopsys Form 8-K Filings (13 February 2024, March 2025 Notes Offering, and 17 July 2025 Closing Report).

Per Synopsys's SEC Form 8-K filings, the cash payout was funded through US$10.0 billion in senior unsecured notes issued in March 2025, a US$4.3 billion term loan draw at closing, and existing balance-sheet cash bolstered by the October 2024 sale of Synopsys's Software Integrity Group to Clearlake Capital and Francisco Partners for up to US$2.1 billion, a carve-out advised by J.P. Morgan.

Taking on US$14.3 billion in new debt moved Synopsys from a net-cash balance sheet to an investment-grade debt profile rated Baa2 by Moody's and BBB by S&P. Both agencies supported those ratings based on the combined company's high free cash flow conversion, driven by recurring subscription revenue and low capital spending, alongside management's pledge to pause share buybacks until prepayable term loans were paid down and gross leverage fell below 2.0 times within twenty-four months.

5. Antitrust Strategy, Remedies, and Downside Scenarios

Clearing antitrust regulators in Washington, Brussels, London, and Beijing was the main hurdle to closing. Rather than fighting prolonged court challenges, the deal teams offered targeted divestitures early in the review process to eliminate direct product overlaps while keeping the core combination intact.

Under consent agreements with the U.S. Federal Trade Commission, the European Commission, and the U.K. Competition and Markets Authority, Synopsys agreed to sell its LightTools and CODE V optical software to Keysight Technologies to resolve overlap with Ansys's Zemax unit, while Ansys agreed to sell its PowerArtist power-analysis software to Keysight to resolve overlap with Synopsys's PrimePower business.

Regulatory Jurisdiction	Clearance Outcome and Structural Condition
United States (FTC)	Cleared via consent order requiring divestiture of optical and RTL power-analysis software to Keysight Technologies
European Union (EC)	Phase I conditional clearance subject to the same global divestiture package to Keysight Technologies
United Kingdom (CMA)	Accepted undertakings in lieu of a Phase 2 reference following the Keysight divestiture agreements
China (SAMR)	Conditional approval granted on 14 July 2025 with behavioral commitments on interoperability and fair supply

Source: U.S. Federal Trade Commission, European Commission, U.K. CMA, and Synopsys Closing Form 8-K Filings.

China's State Administration for Market Regulation granted conditional approval on 14 July 2025 with interoperability and fair-supply commitments, allowing the deal to close on 17 July 2025 ahead of the outside date. Had SAMR blocked the deal or demanded heavier structural divestitures, Synopsys faced a US$1.5 billion reverse termination fee and eighteen months of lost management focus, while a steep drop in Synopsys's stock under the fixed exchange ratio would have reduced headline value for Ansys shareholders.

6. Risks and Post-Close Execution Scorecard

Three risks mattered most after closing. First, product integration: delivering the US$400 million revenue synergy target requires Synopsys's EDA tools and Ansys's physics solvers to work smoothly together in complex customer workflows, a risk reduced by several years of pre-merger technical collaboration. Second, customer pricing: large semiconductor and automotive accounts already spend heavily with both companies and may push back on bundled price increases, though high switching costs and mission-critical software usage support strong renewal rates. Third, leverage: servicing US$14.3 billion of new debt alongside softer demand in China and ongoing R&D investment requires steady free cash flow conversion and post-merger cost discipline.

Post-Close Milestone	Status (Late 2026)	Verified Execution Detail
Regulatory Divestitures	Completed	Optical Solutions Group and PowerArtist sold to Keysight Technologies in October 2025
Deleveraging	On Track	Operating cash flow directed toward prepayable term-loan paydown toward <2.0x gross leverage
Product Integration	Launched	First combined silicon-to-system workflows released for AI hardware customers
Capital Returns	Resumed	US$1.0 billion accelerated share repurchase with JPMorgan Chase announced 5 October 2026

Source: Synopsys Form 8-K Filings (October 2025 Divestiture Closing and 5 October 2026 ASR Agreement).

7. Key M&A Lessons for Deal Practitioners

This transaction serves as a practical reference for how a large strategic software acquisition can be structured with both commercial ambition and balance-sheet discipline.

Deal-Structuring Principle	Takeaway for M&A Teams
Early Structural Remedies	Offering targeted divestitures early can clear multi-jurisdictional antitrust reviews without damaging the core strategic combination.
Share-Issuance Caps	A hard share-issuance cap (19.9999% in this transaction) protects acquirer governance control and avoids a separate buyer shareholder vote.
Credible IG Deleveraging	High-multiple software acquisitions can retain investment-grade ratings when recurring free cash flow backs a clear two-year paydown plan.
Deep Workflow Integration	Cross-selling between adjacent technical software platforms only works when products are deeply integrated rather than merely bundled.

Source: Author synthesis based on Synopsys and Ansys transaction filings.